(15)

Ans:- Generally $A_2 = P\left(1+\frac{R}{100}\right) \Rightarrow P = \frac{A_2}{\left(1+\frac{R}{100}\right)}$

In this question $A_2 = 882$ for 2 yrs.

We want Principle for the 1st yr & 2nd yrs. like

$$P = P_1 + P_2 = \frac{A_2}{\left(1+\frac{R}{100}\right)^1} + \frac{A_2}{\left(1+\frac{R}{100}\right)^2}$$

$$= \frac{882}{1.05} + \frac{882}{(1.05)^2} = \frac{88200}{105} + \frac{8820000}{11025}$$

$P = 840 + 800 = 1640/-$

Ex:-16:- [B.Q:-20]:- A sum of money invested @ CI for 800/- for 3 yrs. & to 840/- for 4 yrs. Find R% p.a ?

Ans:- $A_{2-3y} = 800$, $A_{2-4y} = 840$ to find out 'R'

$$\frac{A_{2-4y}}{A_{2-3y}} = \frac{P(1+R/100)^4}{P(1+R/100)^3} \Rightarrow \frac{840}{800} = 1+\frac{R}{100}$$

$$\Rightarrow 1.05 = 1+\frac{R}{100} \Rightarrow \boxed{R = 5\%}$$

Ex:-17:- [B.Q:-21]:- The least no. of complete yrs in which a sum of money put out @ 10% CI will be more than double Rs ?

Ans:- $A_{2-1} = 1.1P < 2P$

$A_{2-2} = 1.1^2P = 1.21P < 2P$

$A_{2-3} = (1.1)^3P = 1.331P < 2P$

$A_{2-4} = 1.331P + 0.1331P = 1.4641P < 2P$ (10% of)

$A_{2-5} = 1.4641P + 0.1464P = 1.6105 < 2P$

$A_{2-6} = 1.6105P + 0.16105P = 1.77155 < 2P$

$A_{2-7} = 1.77155P + 0.1772P = 1.948 < 2P$

$A_{2-8} = 1.948P + 0.1948P = 2.1428 > 2P$

After [8 yrs].

Ex-18:- [B.Q:-22] :- A sum of money doubles itself @ CI in 10 yrs. In how many yrs will it become 4 times?

Ans:- $P \xrightarrow[+10yrs]{\times 2} 2P \xrightarrow[+10y]{\times 2} 4P$

$P \longrightarrow 4P \longrightarrow$ in $(10+10)$ = 20 yrs

Ex-19:- [B.Q:-23] :- 5200/- was partly invested in scheme A @ 10% p.a CI for 2 yrs & partly in scheme B @ 10% p.a. SI for 4 yrs. Both the schemes earns equal interests. How much was invested in scheme A?

Ans:- CI for 2 yrs in A = SI for 4 yrs in B

lets assume invest in scheme A as x. then

scheme B is $(5200-x)$

$$P\left(\left(1+\frac{R}{100}\right)^n - 1\right) = x\left((1.1)^2 - 1\right) = (10\%+10\%+10\%+10\%) \text{ of } (5200-x)$$

$\Rightarrow 0.21x = 0.4(5200-x)$

$\Rightarrow 21x = 40(5200-x) \Rightarrow 21x+40x = 40\times 5200$

$\Rightarrow 61x = 208000 \Rightarrow x = 3409.83 \cong 3410$

Ex-20:- [B.Q:-25] :- A sum 550/- repaid in 2 equal installments @ 20% p.a in 2 yrs. what is each equal installment?

$$P = P_1 + P_2 = \frac{x}{\left(1+\frac{R}{100}\right)^1} + \frac{x}{\left(1+\frac{R}{100}\right)^2}$$

$$550 = \frac{x}{(1.2)} + \frac{x}{(1.2)^2} = \frac{1.2x+x}{1.44}$$

$\Rightarrow 1.44(550) = 2.2x \Rightarrow \frac{144}{72}(550) = 220x$

$\Rightarrow x = 360/-$